Point-Based Mask Refinement for X-Ray Instance Segmentation

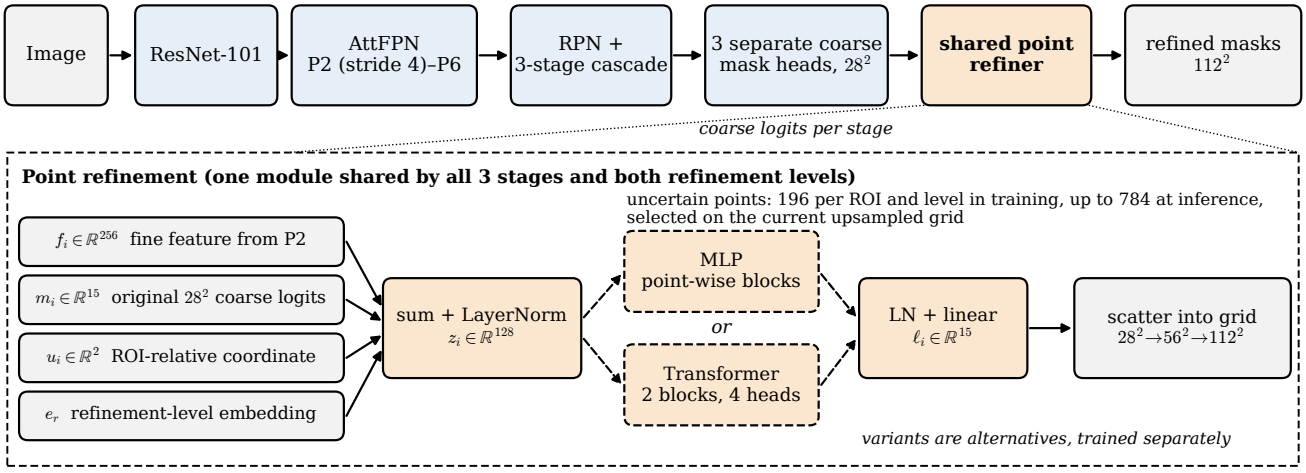


Figure 1: Our framework. Three separate cascade mask heads give coarse 28^2 masks; one shared point refiner updates uncertain locations at 56^2 and 112^2. MLP and transformer are alternative, separately trained variants.

1 Introduction

Prohibited items in X-ray baggage images overlap and are often thin, yet region-based detectors predict each instance mask on a coarse 28×28 grid. We add point-based mask refinement, in the spirit of PointRend [1], to a cascade detector and evaluate it on PIXray [2]. We tested a proposed transformer variant, in which selected points interact, and a matched MLP variant that processes points independently, introduced as a control. In our experiment the MLP variant performed better and is the strongest method we evaluated.

2 Method

Our method starts from a cascade detector that locates objects and predicts a coarse mask for each instance. It revisits selected uncertain locations, combining fine image features, the original coarse-mask predictions and spatial coordinates. A shared refinement head updates those locations at progressively higher resolutions. We evaluate independent point processing with an MLP and interactions between selected points with a transformer (Fig. 1).

The parent is a three-stage Cascade Mask R-CNN [3] with ResNet-101 [4] and the AttFPN neck released with PIDray [5]; its three separate mask heads output 28×28 logits for 15 classes, and P2 denotes the stride-4 AttFPN map. One refinement module is shared by all stages and both levels. Coarse logits are upsampled to 56^2, points are selected by uncertainty on this *current* grid and refined, and the grid is upsampled to 112^2 and refined again. Training uses 196 points per region and level (75% most uncertain from a threefold candidate pool, 25% random); inference uses up to 784. Each point becomes a token

$$z_i = \mathrm{LN}(E_f(f_i) + E_m(m_i) + E_p(u_i) + e_r), \quad z_i \in \mathbb{R}^{128}, \tag{1}$$

where $f_i \in \mathbb{R}^{256}$ is the P2 feature at the point, $m_i \in \mathbb{R}^{15}$ the *original* 28×28 coarse logits there, $u_i \in \mathbb{R}^2$ its ROI-relative coordinate and e_r a learned embedding of level r; E_f is linear, E_m and E_p are two-layer GELU networks. With Z the tokens of one region,

$$\ell_i = W_o\, G(Z)_i + b_o, \quad \ell_i \in \mathbb{R}^{15}, \tag{2}$$

where G is two pre-LayerNorm residual blocks and a final LayerNorm. The transformer variant uses four-head self-attention over tokens of the same region [6]; the MLP variant replaces attention with a point-wise $128 \to 256 \to 128$ MLP, keeping norms, residuals, feed-forward and output layers. The predicted ℓ_i replace the grid values at the selected locations. The refiner adds 458,383 (transformer) or 458,127 (MLP) parameters. Training minimises

$$L = L_{\text{base}} + \sum_{s=1}^{3} w_s \frac{L_{\text{point}}^{s,56} + L_{\text{point}}^{s,112}}{2}, \quad w = (1, 0.5, 0.25), \tag{3}$$

with L_{base} the unchanged detector loss and $L_{\text{point}}^{s,r}$ the binary cross-entropy between the ground-truth class channel of ℓ_i and targets sampled from the transformed ground-truth instance masks. Ground truth is used only for this supervision.

3 Experimental Setup

We use the official PIXray split unchanged: 3,560 training images (10,709 instances), 1,486 test images (4,508 instances), 15 categories; annotations are not modified. All methods train 60 epochs with AdamW (learning rate 10^{-4}, weight decay 0.05), mixed precision, gradient clipping at

Table 1: PIXray official test split (1,486 images, 4,508 instances). Bold indicates the best observed value. Δ denotes the difference from the parent baseline in percentage points. Results use one training seed. Mask AR@100 is from segmentation evaluation. Bilayer: recall-enhanced RPN and auxiliary occluder masks. Occ: SAM ViT-L, auxiliary occluder masks, incomplete checkpoint-to-prediction provenance. Baselines are adapted implementations.

Method	Params (M)	Box AP ↑	Mask AP ↑	Mask AP_{75} ↑	Mask AR@100 ↑	Δ Mask AP
Cascade Mask R-CNN + AttFPN	97.1	80.21	69.12	79.30	72.23	—
Cascade + Bilayer Mask Head	105.2	**80.23**	69.22	79.13	72.41	+0.10 pp ↑
Occ Adaptation	336.7	73.70	61.43	69.07	65.06	−7.70 pp ↓
PIDray R0/R1 Adaptation	98.7	80.12	68.99	79.14	72.14	−0.13 pp ↓
Point Refinement—Transformer (Ours)	97.6	80.01	70.07	78.62	73.16	+0.95 pp ↑
Point Refinement—MLP (Ours)	97.6	80.00	**71.50**	**81.41**	**74.42**	**+2.38 pp ↑**

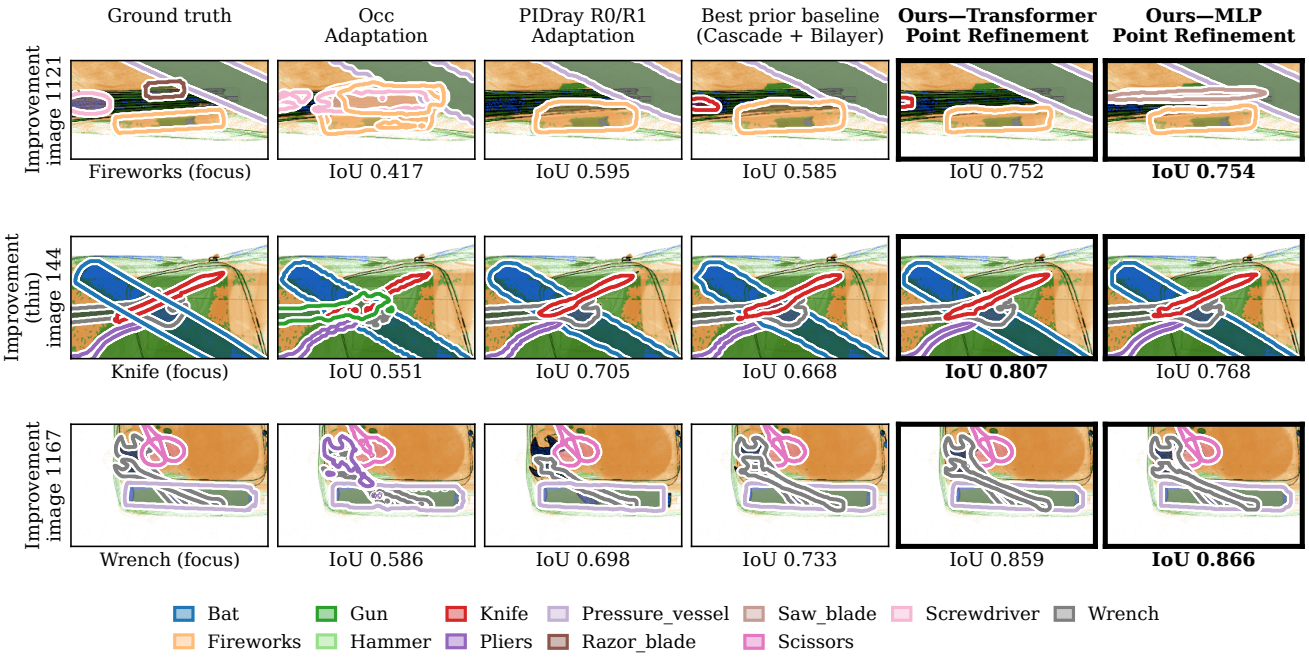


Figure 2: Coloured masks on identical crops of preserved predictions (one colour per category; background tones are X-ray pseudocolour). IoU is each method's best same-category mask IoU with the focus instance; every best mask scores above both the evaluation (0.05) and display (0.30) thresholds, so no panel hides a better mask. Rows are *illustrative improvement cases*, selected where both of our variants exceed all three baselines by the largest margin (row 2 within the four hardest categories); they do not represent average performance. **Fireworks**: ours 0.752 (transformer) and 0.754 (MLP) vs. 0.595 for the best baseline (PIDray); **Knife**: ours 0.807 (transformer) and 0.768 (MLP) vs. 0.705 for the best baseline (PIDray); **Wrench**: ours 0.859 (transformer) and 0.866 (MLP) vs. 0.733 for the best baseline (bilayer).

1.0, batch 2 with 8-step accumulation and seed 42, using flip and random resize (0.5–1.5) with a 512 × 512 crop target. Backbones start from ImageNet with the stem and first residual stage frozen and batch normalisation fixed; all other layers train. The refinement variants differ from the parent only in the ROI head. The final-epoch checkpoint is evaluated without validation-based selection, although the official test split was inspected repeatedly during the project. Evaluation follows COCO [7] (score threshold 0.05, at most 100 detections per image). The comparison methods are adapted implementations: a bilayer mask head [8] with a recall-enhanced RPN, an occlusion-aware bilayer SAM model [9, 10], and the PIDray R0/R1 gate [5] on the parent.

4 Results

The MLP refinement variant achieves the highest observed mask AP among the compared methods, 71.50%, while adding approximately 0.46 million parameters to the parent detector (Table 1). Under the shared PIXray protocol it improves mask AP by +2.38 pp over the parent, +2.28 pp over the bilayer baseline (the previous highest mask AP), +10.08 pp over Occ and +2.51 pp over the PIDray adaptation, and gives the highest mask AP_{75} (+2.11 pp over the parent) and mask AR@100 (+2.19 pp). The gain grows from +0.39 pp at IoU 0.50 to +5.00 pp at IoU 0.95, so it comes from tighter masks. Box AP did not improve (−0.21 pp MLP, −0.20 pp transformer). The transformer variant improves mask AP

Table 2: Paired differences (pp) on GT instances matched by *both* methods (count, % of GT); score ≥ 0.5, one-to-one same-class box IoU ≥ 0.5. Dice $= 2|P \cap G|/(|P| + |G|)$; boundary IoU [11] uses a band of 2% of the image diagonal at original resolution. Impr.: % of pairs with higher IoU.

Pair	Both (%)	ΔIoU	ΔDice	ΔBIoU	Impr.
MLP vs parent	4159 (92.3)	+1.07	+0.65	+1.17	70.8
Transformer vs parent	4177 (92.7)	+0.39	+0.20	+0.47	61.2
MLP vs Transformer	4161 (92.3)	+0.68	+0.45	+0.69	60.9
MLP vs bilayer	4148 (92.0)	+1.12	+0.68	+1.24	71.6

by +0.95 pp over the parent; the MLP variant outperforms the matched transformer variant in this run by +1.43 pp and at all 10 IoU thresholds.

5 Error Analysis

On ground-truth instances matched by both methods (Table 2), the MLP variant raises mean IoU by +1.07 pp and boundary IoU by +1.17 pp over the parent, improving 70.8% of instances; against the bilayer baseline the figures are +1.12 pp, +1.24 pp and 71.6%. Boundary gains exceed region gains in every pair, consistent with the strict-IoU AP improvement; Fig. 2 illustrates more complete fireworks, knife and wrench masks from both our variants than from any baseline.

Refinement does not repair detection: box AP does not improve, and the four hardest categories (Dart, Saw_blade, Knife and Screwdriver) average 51.71% mask AP against 48.74% for the parent (+2.96 pp), with Dart weakest at 36.87%. When an instance is missed or misclassified, there is no mask for the refiner to correct. The MLP control also tests the original hypothesis: under this protocol, point-to-point attention added no accuracy over point-wise refinement with the same inputs (paired IoU +0.68 pp in favour of the MLP). One seed cannot show that attention is generally unhelpful.

6 Conclusion

Point-based refinement of cascade masks from fine features, original coarse logits and coordinates improves segmentation on PIXray. Our MLP variant reaches the highest observed mask AP, 71.50% (+2.38 pp over its parent), with gains concentrated at strict IoU and no box-AP gain, and outperforms the transformer variant in this run.

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